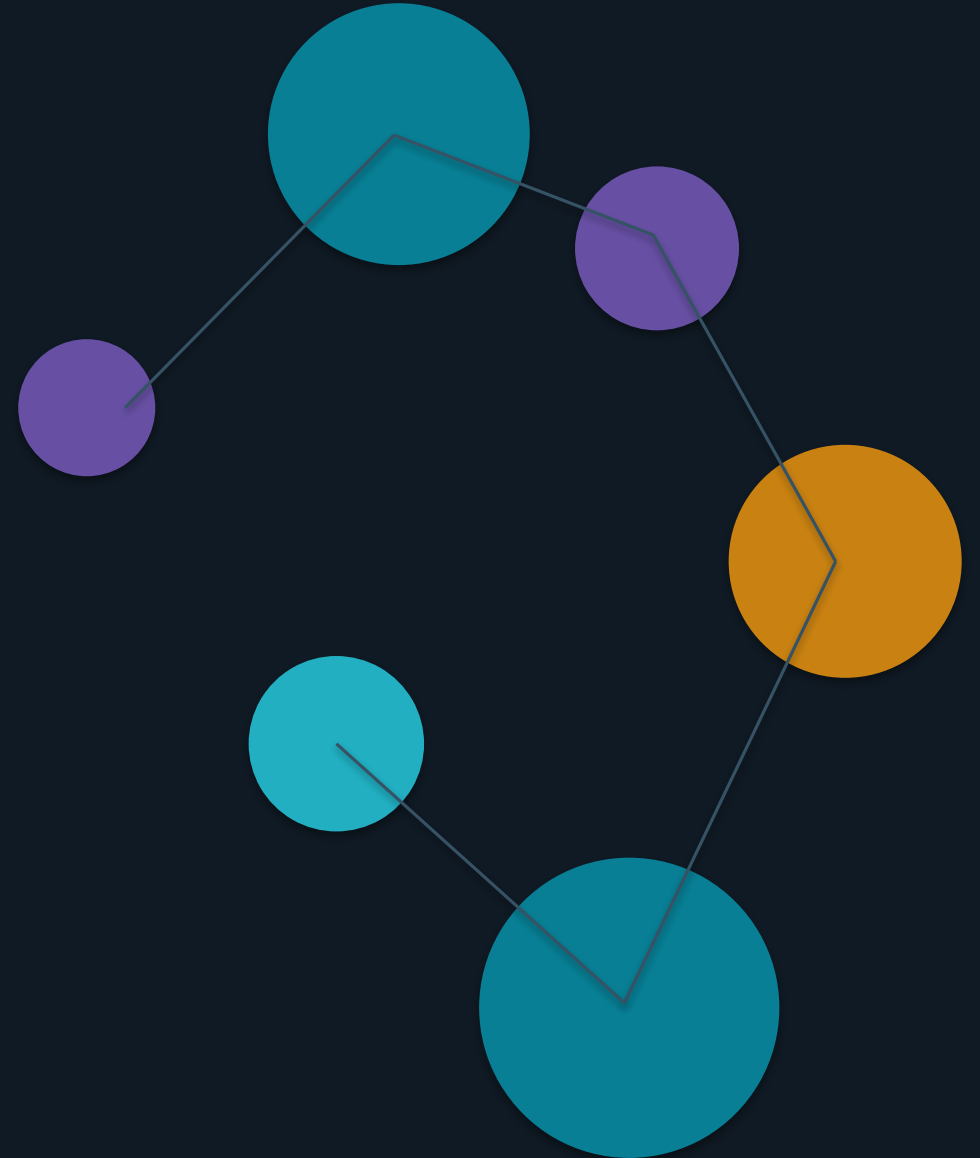


From research question to auditable simulation evidence

A practical research cockpit combining literature memory, constrained LLM reasoning, reproducible Modelica experiments, and claim-level provenance.

MODELICA × LLM-GUIDED RESEARCH × AUDITABLE EVIDENCE

17 AUGUST 2026



The platform is a research control plane—not an autonomous scientist

It accelerates the tractable parts of computational research while making human choices and model limitations visible.

01

What it automates

- Literature retrieval and screening
- Structured gap and hypothesis drafts
- Model selection and bounded parameterization
- Batch execution, analysis, and reporting

02

What remains human

- Choice of research direction
- Approval of the scientific plan
- Interpretation of trade-offs
- Judgment about validity and contribution

03

What is auditable

- Frozen literature snapshot
- Approved plan and protocol hashes
- Pinned runtime and raw-file fingerprints
- Revised analysis and claim links

Core proposition: move faster without collapsing generation, execution, evidence, and scientific judgment into one opaque step.

Computational research is fragmented at exactly the points that matter

The costly work is not one simulation—it is preserving intent and evidence across the whole loop.

1

Literature drift

Searches change; rationales disappear; missing abstracts are easily overinterpreted.

2

Plan drift

Parameters and metrics change between the research question, execution, and analysis.

3

Environment drift

Model libraries, solvers, and runtime images silently alter behavior.

4

Evidence drift

Final prose becomes detached from cases, files, comparisons, and caveats.

The design response

Treat every transition as a typed, inspectable artifact—not just a prompt or log line.

Scope is intentionally narrow: defensible simulation studies

The current domain is data-center cooling, but the control pattern is designed to generalize.

IN SCOPE

- Turn a research profile into ranked, traceable hypothesis candidates
- Compile approved ideas into registered Modelica experiments
- Execute matched studies inside a pinned OpenModelica environment
- Verify raw results, analyze under a preregistered protocol, and draft linked claims

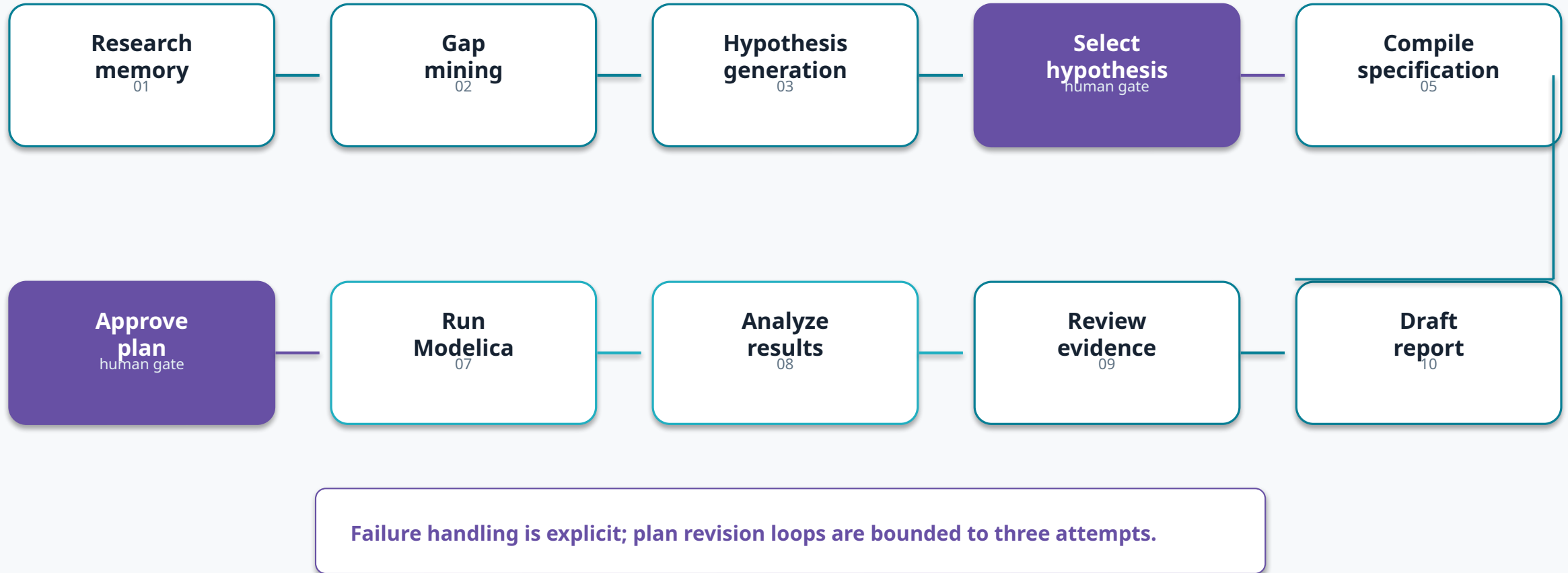
NOT CLAIMED

- Autonomous novelty determination or publication readiness
- Physical validation without calibration data
- Universal correctness of an LLM-generated hypothesis
- Replacement for a PI, domain expert, or independent reviewer

Success means a shorter path to a reviewable study—not an automated scientific verdict.

One closed loop connects intent, execution, and evidence

Eleven workflow nodes create explicit hand-offs; two human gates stop silent scientific drift.



Four layers separate probabilistic reasoning from deterministic evidence

This boundary is the core reliability mechanism.

01**01 · RESEARCH INTERFACE**

Profiles · gates · run cockpit · evidence review

02**02 · AGENT WORKFLOW**

Typed state · constrained LLM calls · revision loops

03**03 · SCIENTIFIC RUNTIME**

Registry · compiler · OpenModelica · batch analysis

04**04 · EVIDENCE STORE**

Snapshots · raw series · hashes · revisions · exports

Boundary rule**LLMs may propose.****Schemas must validate.****Compilers bound.****Runtimes measure.****Hashes prove.**

A fluent answer is never treated as a simulation result.

Constrained intelligence, not free-form automation

The LLM is useful where ambiguity is real—synthesis, framing, and drafting. Deterministic components take over where correctness can be checked.

Research Memory

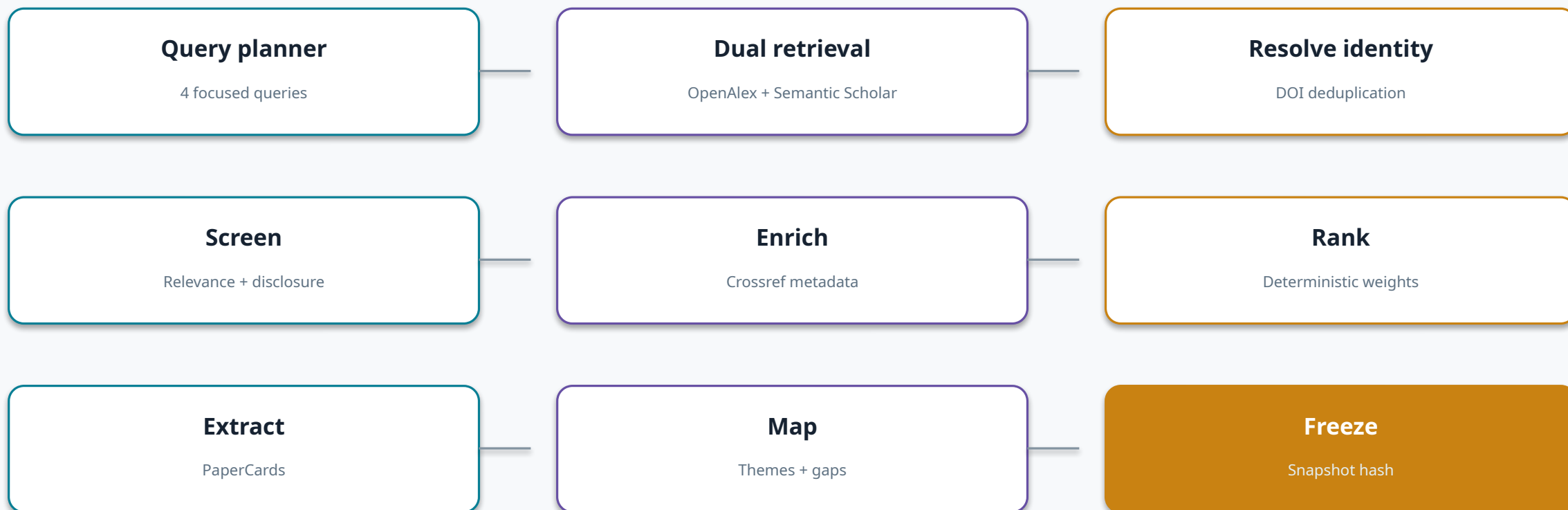
Schema constraints

Human gates

Deterministic compiler

Research Memory freezes the literature context used by the run

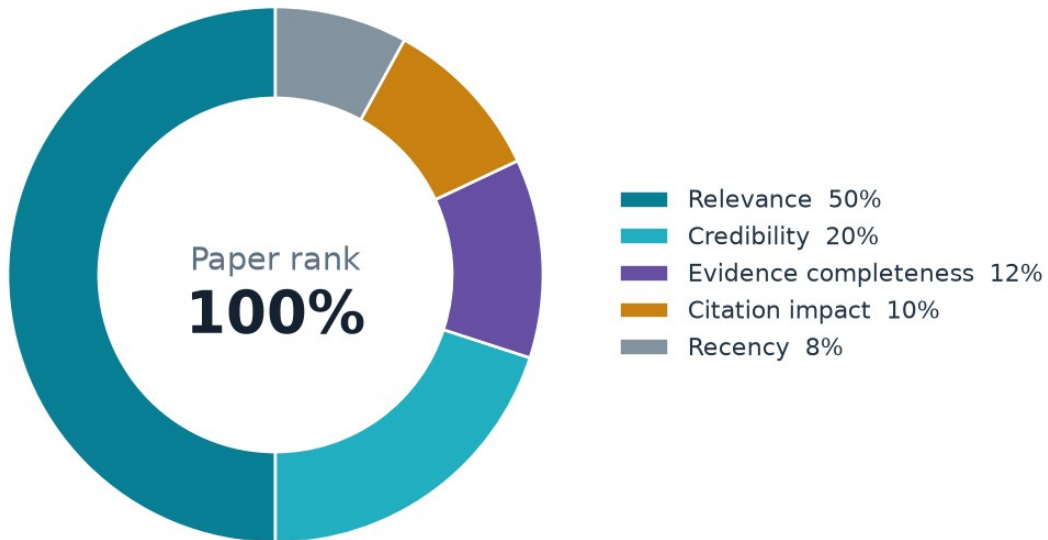
A reproducible snapshot makes later hypotheses and claims inspectable.



Missing abstracts are marked insufficient; the platform does not invent methods or limitations to complete a card.

Ranking is deterministic and credibility is disclosed

Relevance dominates the score; venue status and evidence completeness remain visible rather than silently blended.



Verified

Cross-source identity and publication metadata align.



Likely / uncertain

Evidence is usable with an explicit confidence label.



Preprint

Not discarded—but never presented as peer-reviewed.

The ranking can prioritize attention; it cannot determine truth.

The LLM layer is replaceable, constrained, and measured

Current default: DeepSeek V4 Flash through an OpenRouter-compatible endpoint.

0.2

Temperature

Bias toward repeatable structured outputs

JSON

Pydantic schemas

Reject malformed or incomplete responses

Retry

Validation repair

A bounded second chance—not silent coercion

Usage

Token capture

Per-call accounting retained in run state

Provider-agnostic

OpenAI-compatible client boundary allows model replacement without changing scientific artifacts.

Reasoning separated

Generated hypotheses and prose never overwrite observed Modelica outputs or hashes.

Failure visible

Provider warnings, schema failures, and missing evidence propagate to the user interface.

Model choice is a configuration decision; evidentiary rules are platform invariants.

Every task is assigned to the least ambiguous mechanism

This division of labor reduces hallucination surface and makes failures diagnosable.

LLM-SUITABLE

- 1 Synthesize literature themes**
Useful where judgment and language matter.
- 2 Propose research gaps**
Useful where judgment and language matter.
- 3 Draft candidate hypotheses**
Useful where judgment and language matter.
- 4 Explain results in plain language**
Useful where judgment and language matter.

DETERMINISTIC

- 1 Resolve and rank papers**
Used where the rule can be made explicit and tested.
- 2 Validate schemas and bounds**
Used where the rule can be made explicit and tested.
- 3 Compile experiment cases**
Used where the rule can be made explicit and tested.
- 4 Compute metrics, hashes, and quality gates**
Used where the rule can be made explicit and tested.

Human judgment spans both columns: selecting questions, approving plans, and interpreting evidence.

Typed artifacts are the hand-offs between agents

Each stage receives a bounded object rather than an unstructured transcript.

01

ThesisProfile

domain · interests · objectives · constraints

02

PaperCard

identity · methods · findings · limitations

03

Hypothesis

question · mechanism · contribution · risks

04

ExperimentPlan

cases · pairs · metrics · protocol

05

ResultBundle

files · fingerprints · metrics · warnings

06

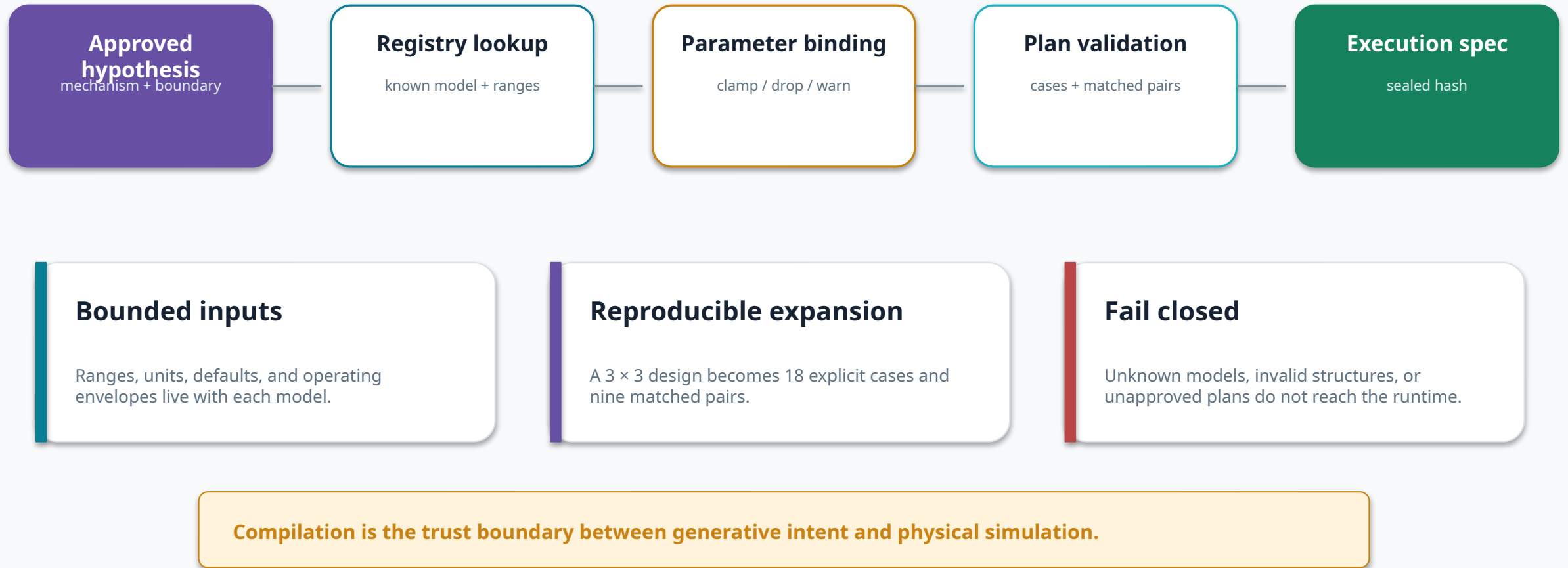
Claim

statement · comparison IDs · paper IDs

Typed state makes a run resumable, testable, and reviewable without replaying the LLM conversation.

The scientific compiler turns an approved idea into executable cases

It selects only registered models and deterministically clamps or rejects unsafe parameters.



Two human gates preserve scientific agency

The platform can operate quickly without treating automation as authorization.

H1

Gate 1 · Select hypothesis

Reviewer sees ranked candidates, the research question, novelty claim, feasibility, supporting papers, risks, model fit, provider warnings, and ranking rationale.

H2

Gate 2 · Approve experiment plan

Reviewer sees the model, parameters, operating ranges, cases, matched comparisons, outputs, metrics, protocol, validation issues, and disclosures before any compute begins.

Rejecting a plan creates a visible revision request; the loop is capped at three revisions.

Modelica is the executable theory layer

Equation-based models make assumptions, state, and conservation relationships explicit—while the registry prevents ‘any model, any parameter’ execution.

[OpenModelica](#)[Model registry](#)[Credibility tiers](#)[Pinned libraries](#)

Why Modelica fits this research workflow

It supplies an inspectable physical model beneath the language-model layer.



Declarative physics

Components describe equations and connections rather than a fixed procedural solver sequence. That is valuable for multi-domain thermal and fluid systems.



Composable systems

Rooms, heat exchangers, pumps, chillers, controllers, weather, and sensors can be assembled while retaining their physical interfaces.



Reproducible execution

OpenModelica compiles each sealed specification inside a pinned image; outputs become files that can be fingerprinted and reanalyzed.

Modelica increases transparency of the model—but does not by itself validate the model against reality.

The model registry carries credibility and operating boundaries

Models are not interchangeable; each exposes its own inputs, outputs, validation tier, and limitations.



DataCenterRoom

Conceptual · v1.0.0

Single-zone thermal balance with proportional cooling and constant COP. Fast enough for interactive hypothesis exploration.

EXPLORATORY



Non-integrated plant

Library-validated · Buildings 13.0.0

Pinned wrapper around the non-integrated primary-secondary waterside-economizer example. Baseline in the milestone study.

LIBRARY REGRESSION



Integrated plant

Library-validated · Buildings 13.0.0

Pinned wrapper around the integrated primary-secondary waterside-economizer example. Candidate configuration.

LIBRARY REGRESSION

DataCenterRoom is a transparent conceptual model

Its value is speed and interpretability—not facility-level fidelity.

ROOM ENERGY BALANCE

$$C \cdot dT/dt = Q_{IT} + UA(T_{amb} - T) - Q_{cool}$$

$$Q_{cool} = \text{clamp}[kp(T - T_{set}), 0, Q_{max}]$$

$$P_{cool} = Q_{cool} / COP$$

8 bounded parameters

0 W

energy-balance residual

26.818 °C

analytic equilibrium

100 °C

saturation branch equilibrium

bitwise

repeatability

passed

unit consistency

directional

load/capacity sensitivity

Omitted: spatial hot spots, humidity, airflow network, equipment transients, sensor uncertainty, and calibration data.

The milestone benchmark compares two Buildings library configurations

Both share the same room load, weather profile, outputs, horizon, and chilled-water setpoint.

NON-INTEGRATED

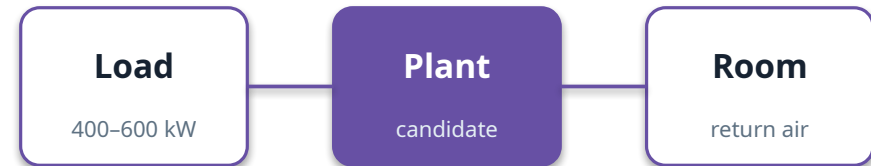
Primary-secondary plant with waterside economizer



Separate non-integrated economizer arrangement

INTEGRATED

Primary-secondary plant with integrated economizer



Economizer integrated into the chilled-water path

Validation tier: agreement with pinned upstream library behavior—not calibration against a physical data center.

A simulation is only reproducible if its environment is named

The manifest records exact solver, libraries, image, models, inputs, code state, and artifacts.

OPENMODELICA

1.26.3

MODELICA STANDARD LIBRARY

4.1.0

BUILDINGS

13.0.0

RUNTIME IMAGE

rcp-openmodelica-buildings

Reproducibility manifest also records the repository revision and dirty state—because uncommitted code is part of the environment.

Artifacts are exported as a ZIP with SHA-256 fingerprints and byte counts.

A preregistered 18-case Modelica benchmark

The study asks whether integrating a primary–secondary waterside economizer improves energy or thermal performance across load and chilled-water setpoint.

18 cases

9 matched pairs

24 h each

8 primary metrics

The design is a complete 3×3 matched comparison

Every operating point runs the non-integrated baseline and integrated candidate under identical declared conditions.



The analysis protocol defines metrics before results are seen

Primary metrics, directionality, units, plausibility ranges, and quality rules are part of the approved plan.

EHVAC

lower

HVAC energy

kWh

PUE

lower

Power usage effectiveness

—

T_{peak}

lower

Peak room temperature

°C

Exceed.

lower

Degree-hours above 27 °C

°C · h

Free

context

Free-cooling duration

h

Partial

context

Partial-mechanical duration

h

Full

context

Full-mechanical duration

h

Switches

lower

Cooling-mode transitions

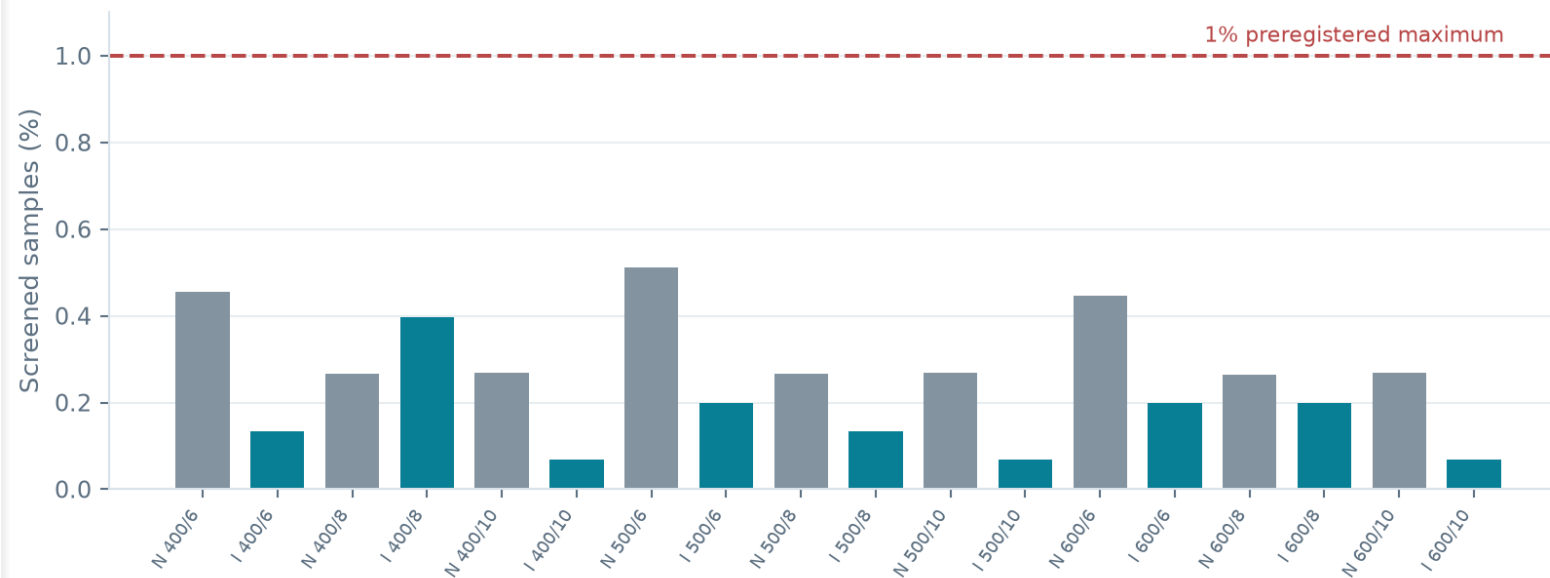
count

Protocol hash 653a8df166...a997e6

A numerical impulse exposed why safeguards matter

OpenModelica trajectories contain isolated nonphysical mover-power events; raw cumulative energy is preserved, while the approved primary metric bridges screened samples.

Numerical-event screening stays below the approved maximum



Physical screen

$$0 \leq \text{PHVAC} \leq 2 \text{ MW}$$

Repair rule

Linear bridge internally; nearest valid value at a boundary.

Observed

0.068–0.512% screened

Raw upstream energy remains available as a diagnostic; it is never overwritten.

The study passes its preregistered quality gate—with one disclosed warning

Quality checks test the study artifact before any conclusion is drafted.

18 / 18

cases successful

9 / 9

matched pairs

72 / 72

comparisons

8+1

passed + warning



Primary metrics complete and finite



Raw-series SHA-256 integrity verified



PUE remains inside 1.0–3.0



Cooling-mode time balances to 24 h



IT energy is monotonic with load

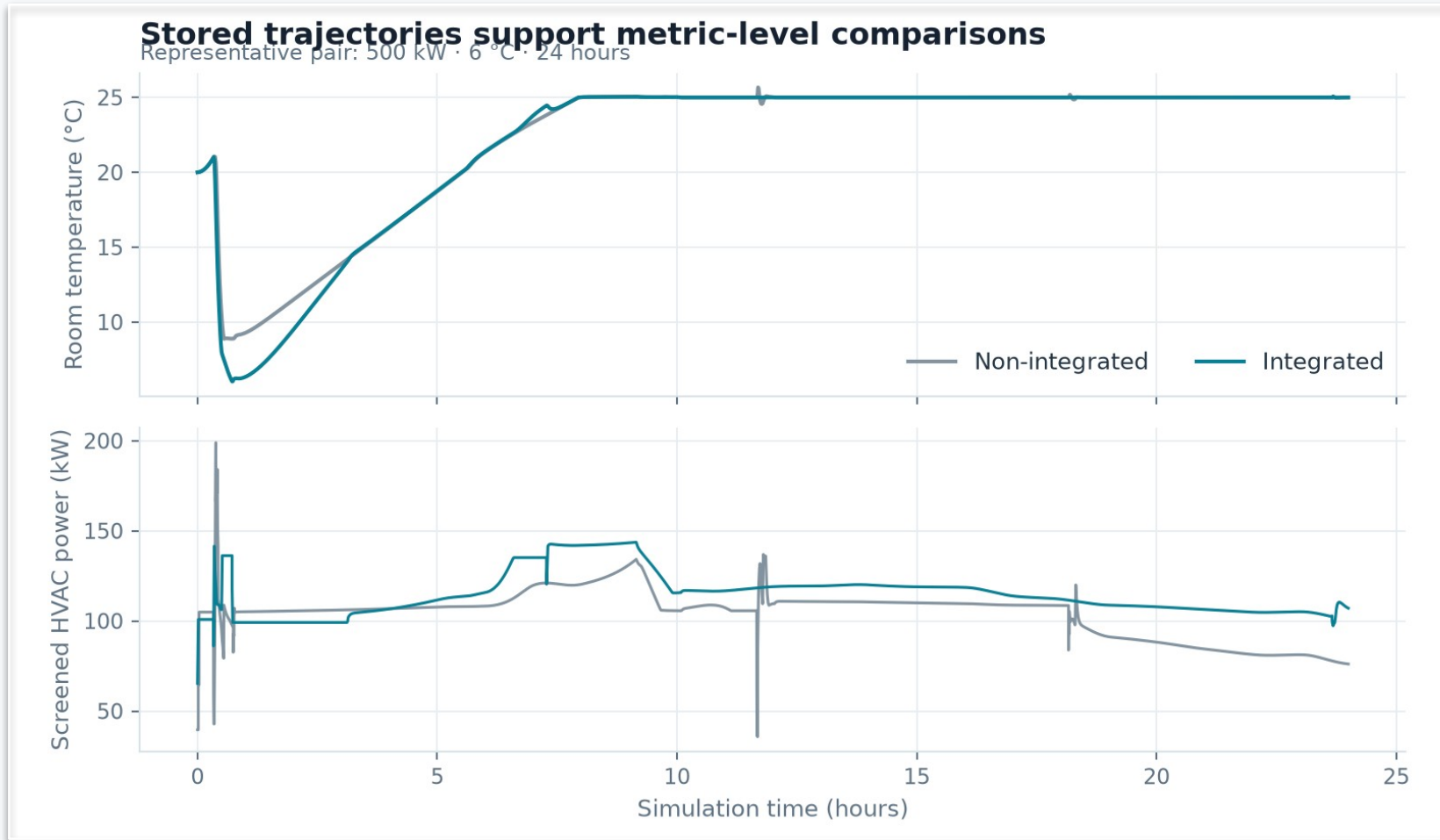


Power screening stays below 1% maximum

Quality-valid means the declared analysis is internally defensible—not that the physical model is calibrated.

Metrics remain connected to the stored trajectories

A reviewer can move from a result summary back to the screened power and room-temperature time series.



Representative pair

500 kW · 6 °C · 24 h

Baseline

Non-integrated plant

Candidate

Integrated plant

The integrated configuration is not an energy winner

Across all nine operating points it uses more screened HVAC energy. At 6 °C it also maintains a cooler peak room temperature—revealing a trade-off worth investigating.

+10.3% mean energy

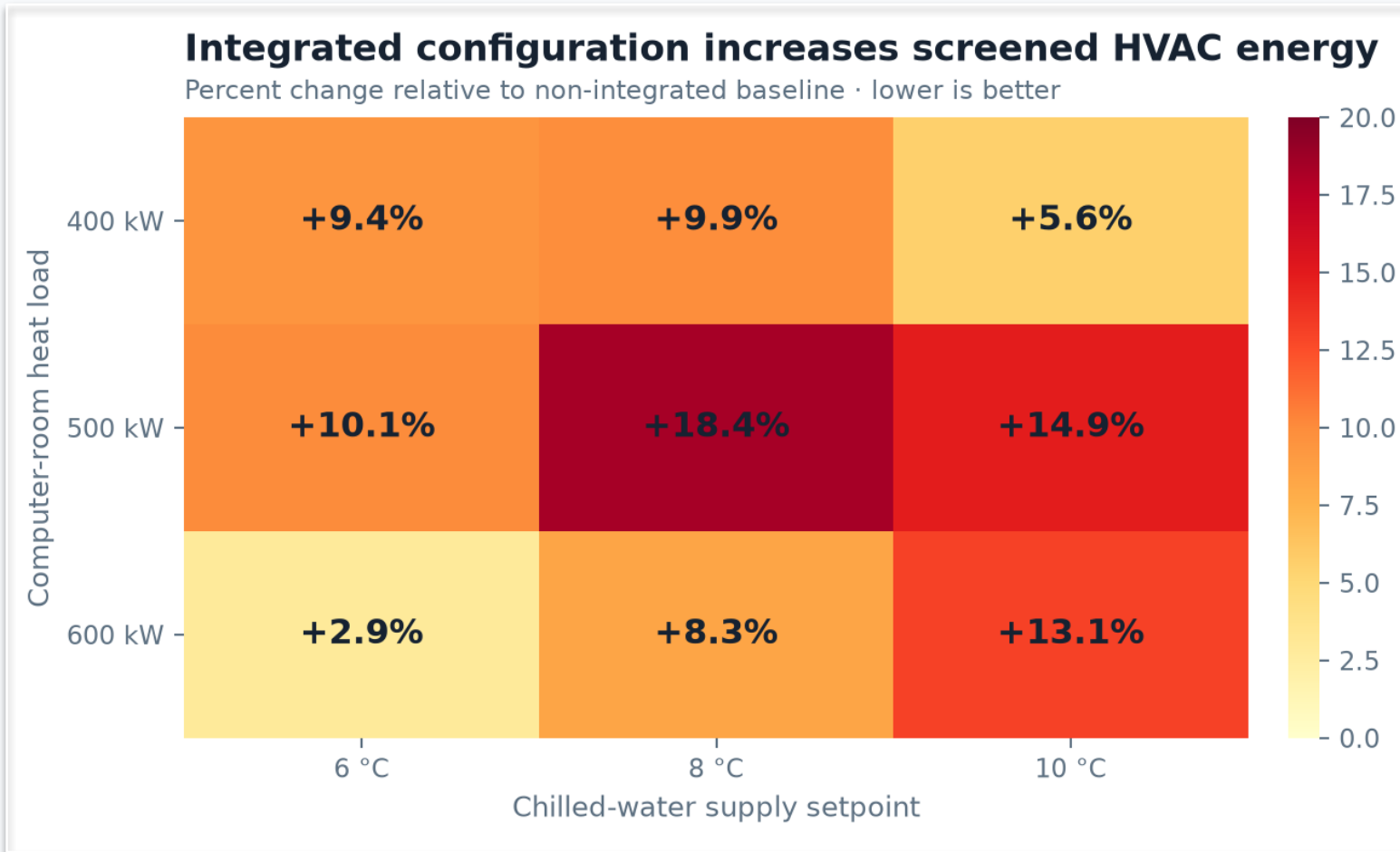
0 / 9 energy wins

cooler at 6 °C

0 thermal exceedance

Integrated HVAC energy increases in every matched pair

The increase ranges from +2.9% to +18.4%; mean change is +10.3%.



+10.3%

mean HVAC energy

0 / 9

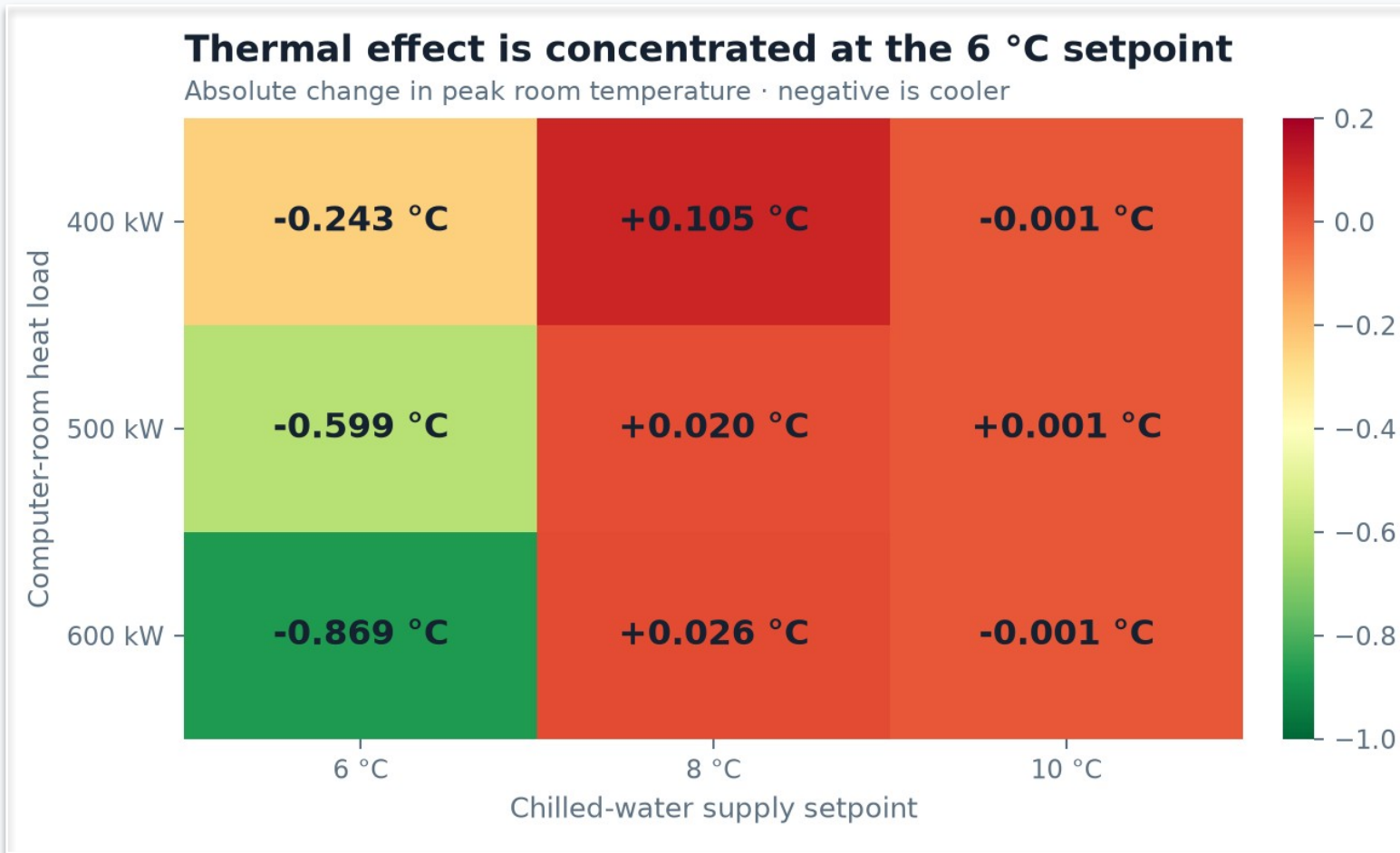
candidate energy wins

+2.9–18.4%

observed range

The thermal benefit appears only at the lowest setpoint

At 6 °C, integration lowers peak room temperature by 0.243–0.869 °C; at 8–10 °C the effect is negligible or slightly adverse.



-0.869 °C

largest peak reduction

0 °C·h

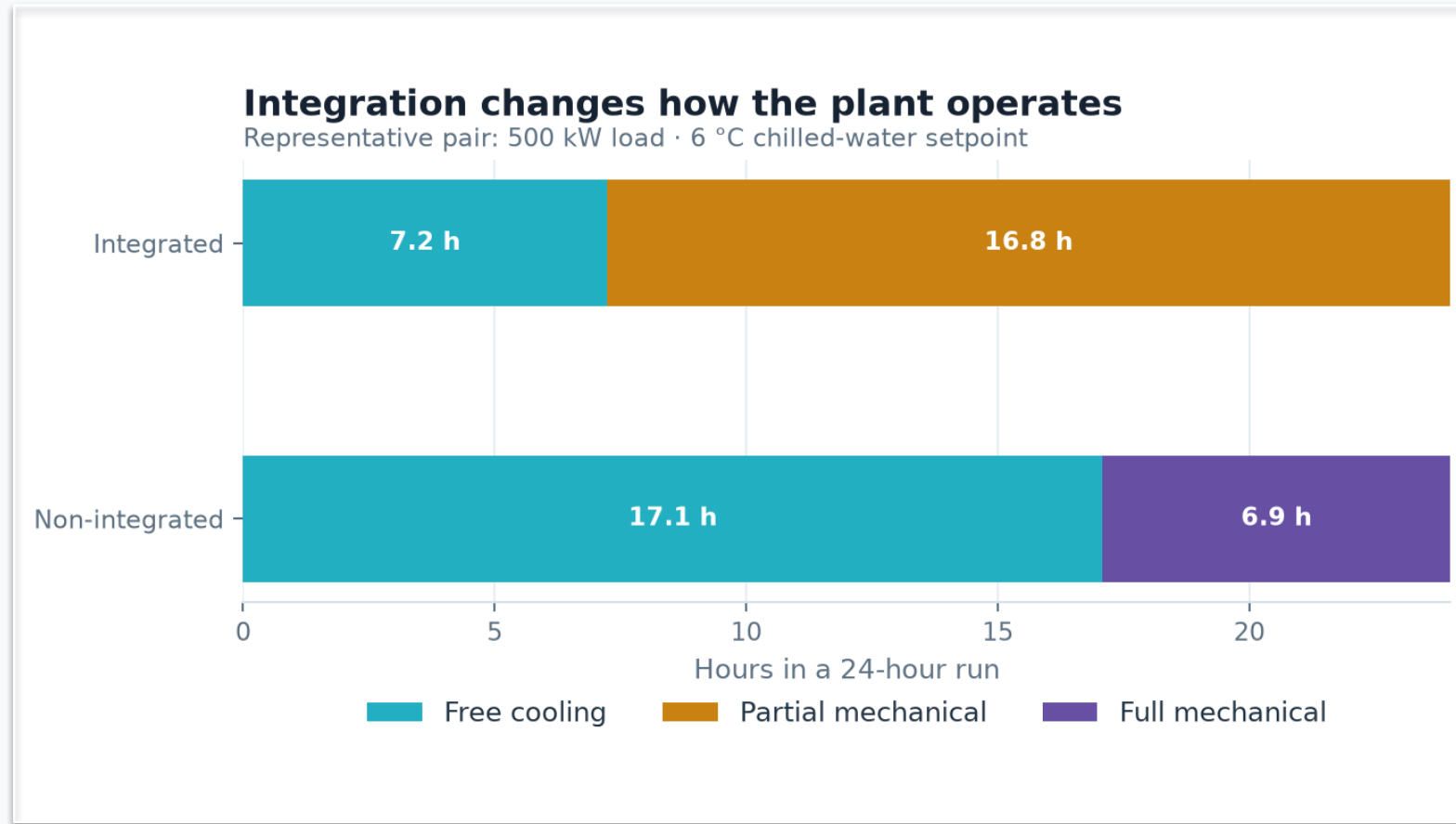
exceedance above 27 °C

+0.105 °C

largest slight worsening

Integration changes operating mode—not simply efficiency

At 500 kW and 6 °C, full-mechanical hours disappear, but partial-mechanical operation expands to 16.8 hours.



Baseline

17.1 h free + 6.9 h full

Integrated

7.2 h free + 16.8 h partial

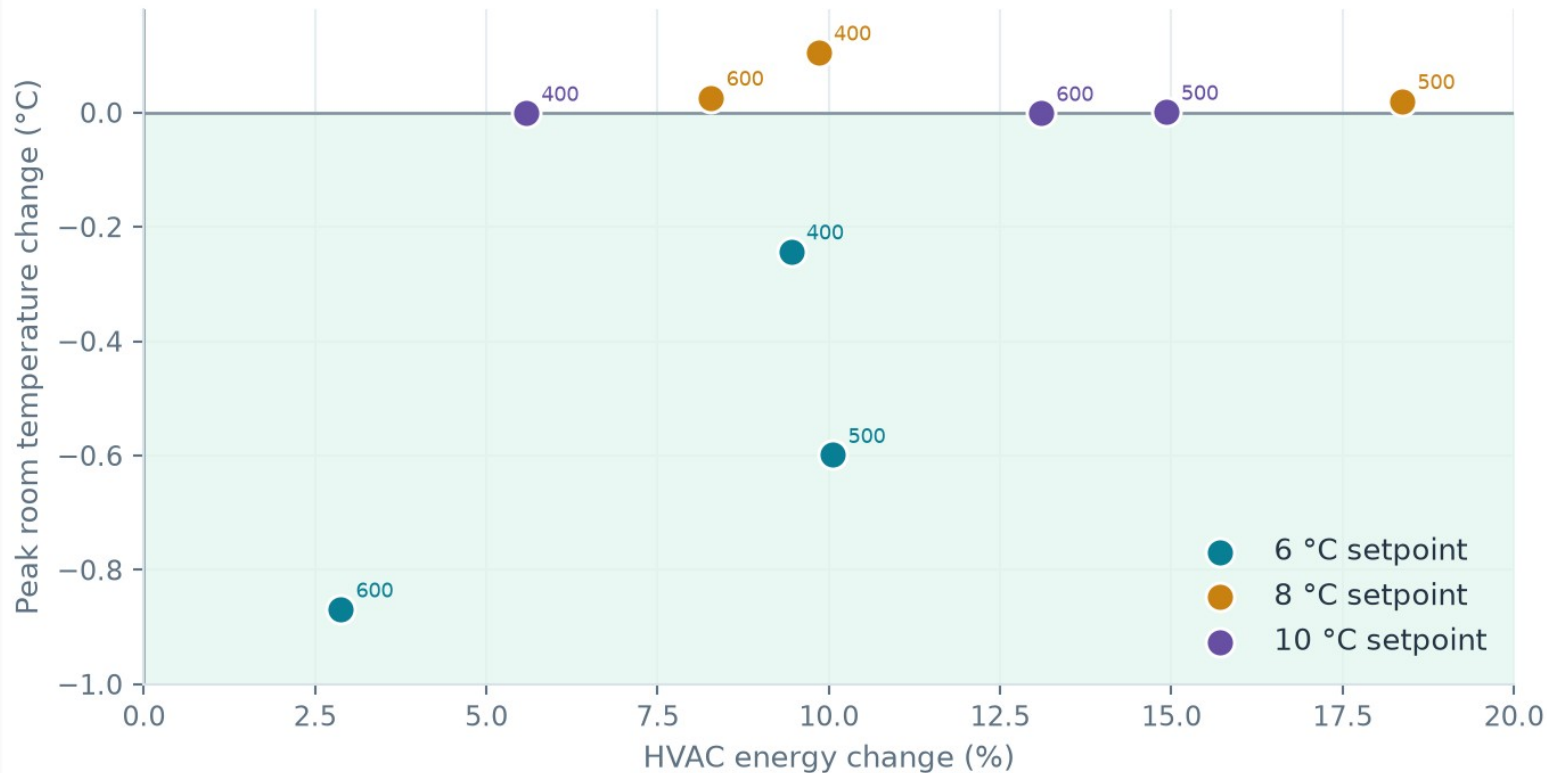
Both

4 mode switches

The correct conclusion is conditional—not 'integrated is better'

The data suggest an energy-thermal-margin trade-off at 6 °C and no energy advantage elsewhere.

The benchmark reveals a trade-off—not a winner



Supported

Integration can reduce peak temperature at 6 °C, with higher HVAC energy.

Not supported

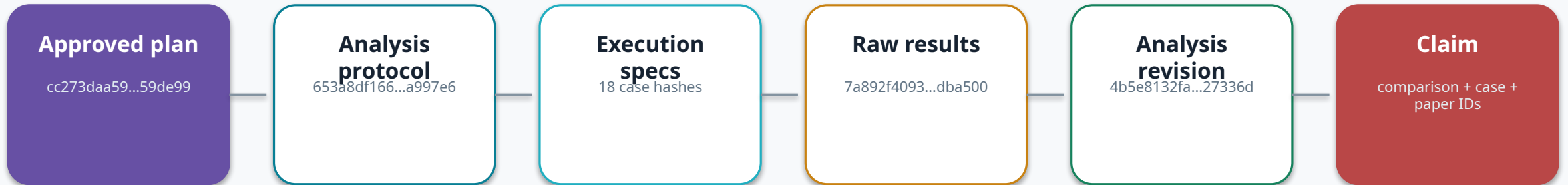
Integration minimizes energy across the evaluated domain.

Next question

What control objective values thermal margin enough to justify the cost?

Every conclusion can be traced back to a sealed chain

The same chain supports review, reanalysis, and export without trusting the narrative layer.



Tamper evidence

Raw files are checked against recorded SHA-256 fingerprints before analysis.

Append-only analysis

A protocol change creates a new revision; the previous result is retained.

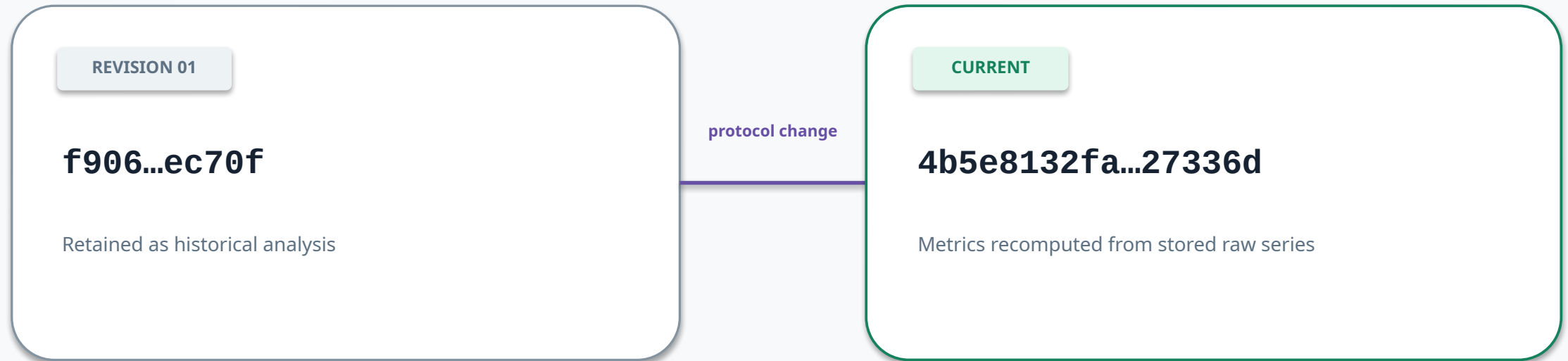
Claim links

Draft statements cite comparison, case, and supporting-paper identifiers.

The report is a view over evidence—not the evidence itself.

Reanalysis is versioned instead of silently replacing history

The benchmark currently has two stored analysis revisions; the newest records that the protocol changed.



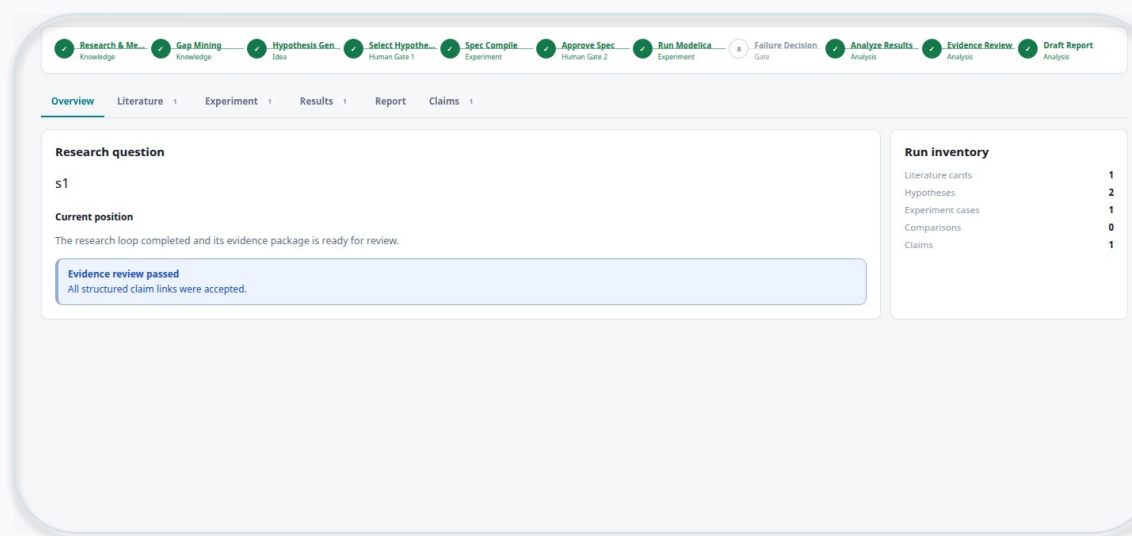
Result: methodological improvement without erasing the record that supported earlier outputs.

The interface is designed around reviewable research state

Light and dark themes, keyboard-visible controls, responsive layouts, URL-addressable views, and explicit lifecycle actions are now implemented.

RUN	TOPIC	STATUS	CURRENT STAGE	CREATED	ELAPSED	
660b69fb	reproducible	done	Complete	45m ago	01s	Archive
a852c34e	Cooling saturation resilience	done	Complete	45m ago	02s	Archive
6b736c96	editable report	done	Complete	45m ago	01s	Archive
675f8a04	t	done	Complete	45m ago	01s	Archive
4887f3a5	gated run	done	Complete	45m ago	01s	Archive
73c702fc	test topic	done	Complete	45m ago	00s	Archive
a2dc8720	reproducible	done	Complete	58m ago	01s	Archive
e1cd20f4	Cooling saturation resilience	done	Complete	58m ago	01s	Archive
61fc167b	editable report	done	Complete	58m ago	01s	Archive

RUN PORTFOLIO



RUN COCKPIT

Status at a glance

Active, attention, completed, and failed runs remain visible.

Lifecycle navigation

Stages, gates, tabs, claims, and artifacts are one click away.

Safe operations

Archive and retry update the same view after the backend succeeds.

Researchers can move from profile to literature to simulation

The UI exposes the same boundaries as the scientific workflow rather than hiding them behind a chat box.

Runs Knowledge Models

Draft restored
Your unfinished run configuration was recovered from this browser tab. Dismiss

1 Research profile 2 Ranked ideas 3 Run configuration 4 Confirmation

Build a research profile

Combine your research intent with the model registry and literature scan to rank feasible thesis directions.

Enter topic directly

Domain
Data-center digital twins

Interests (comma separated)
cooling optimization, GPU rooms

Objectives
reduce cooling energy, quantify resilience

Preferred methods
Modelica simulation

Thesis constraints
12-week timeline, no facility data

Experience level
Intermediate

Generate five thesis ideas ...

01 · Frame

Research profile + constraints

Runs Knowledge Models

Search titles, tags, methods...

All credibility All publication types

energy efficiency 0 data center 1 cooling optimization 4 deep reinforcement learning 3 actor-critic 2 data center energy efficiency 3 reinforcement learning 1

data center cooling 1 ddpg 1 cooling systems 1 ict processes 1 systematic review 1

REVIEW UNCERTAIN unknown 0.0 2018

Air Flow Measurement and Management for Improving Cooling and Energy Efficiency in Raised-Floor Data Centers: A Survey

IEEE Access - 53 citations - extraction from title

PROBLEM Improving cooling efficiency and energy efficiency in raised-floor data centers, which face challenges from increasing power density and scale, leading to over-provisioning of cooling resources and high energy costs.

LIMITATIONS Survey paper; no original experimental validation; Focuses only on air cooling, not other cooling technologies; Limited to raised-floor data center configurations

Why selected

data center cooling air flow management raised-floor data center energy efficiency cooling efficiency thermal measurement

REVIEW UNCERTAIN unknown 0.0 2019

DeepEE: Joint Optimization of Job Scheduling and Cooling Control for Data Center Energy Efficiency Using Deep Reinforcement Learning

Unknown venue - 91 citations - extraction from title

PROBLEM Joint optimization of job scheduling and cooling control for data center energy efficiency, addressing the failure of prior approaches to capture system dynamics and the complexity of state and action spaces.

LIMITATIONS Simulation-based evaluation may not capture real-world dynamics fully; Requires validation in real data centers; Training complexity of deep reinforcement learning algorithms; Generalizability to different data center configurations and workloads

Why selected

deep reinforcement learning job scheduling scaling control data center energy efficiency parameterized action space two-time scale control

REVIEW UNCERTAIN unknown 0.0 2021

Improving Data Center Energy Efficiency Through End-to-End Cooling Modeling and Optimization: Final Report

Unknown venue - 2 citations - extraction from title

PROBLEM Improving data center energy efficiency through end-to-end cooling modeling and optimization, specifically by optimizing cooling system operation, airflow management, and their simultaneous optimization.

LIMITATIONS Results are based on specific partner data centers and may not generalize; Savings depend on specific energy-saving measures employed; Hypothetical simple data center shows lower savings (6.8%); Software features may require specific retrofitting or conditions

Why selected

REVIEW UNCERTAIN unknown 0.0 2023

Interpretable machine learning for building energy management: A state-of-the-art review

Advances in Applied Energy - 313 citations - extraction from title

PROBLEM Black-box nature of machine learning models hinders user understanding and trust in building energy management.

LIMITATIONS Different terminologies cause confusion about model interpretability; Performance of interpretable ML across tasks is difficult to compare; Current techniques like SHAP and LIME provide only limited interpretability.

Why selected

REVIEW UNCERTAIN unknown 0.0

Reinforcement learning for data efficiency optimization: A system review and research roadmap

Applied Energy - 40 citations - extraction from title

PROBLEM Optimization of data center particularly cooling system using reinforcement learning.

LIMITATIONS Lack of real-time validation; Absence of metrics for reporting improvements

Why selected

02 · Ground

Credibility-aware PaperCards

Runs Knowledge Models

Registered model
DataCenterRoom

Simulations
Runs and run a controlled single-case simulation.

Conceptual v1.0.0-conceptual

If a data center room: IT heat load, envelope losses, proportional cooling with COP-based electric power: Fast, steady-state simulation.

Q_it 1000-500000 W

C_room 100000-100000000 W/K

UA 0-50000 W/K

T_amb -20-55 degC

T_set 15-35 degC

Q_cool_max 1000-1000000 W

k_p 100-1000000 W/K

COP 1-10

Stop time (seconds)
86400

Run simulation

	DEFAULT	UNIT	RANGE
Q_it	50000	W	1000-500000
C_room	2000000	J/K	100000-100000000
UA	2000	W/K	0-50000
T_amb	30	degC	-20-55
T_set	24	degC	15-35
Q_cool_max	800000	W	1000-1000000
k_p	20000	W/K	100-1000000
COP	3.5	-	1-10

03 · Explore

Bounded model quick simulation

Reviewers reach the platform over an authenticated link

One supervised process on loopback, an outbound TLS tunnel, and HTTP Basic auth covering the API, the SPA, and the live run stream alike.



Fail closed by default

`rcp serve` will not start without a password, will not accept one under 16 characters, and will not disable auth on a non-loopback host.

Nothing streamed was broken

A pass-through gate, not a response wrapper: run events keep flowing (with a 15 s keepalive for proxy idle timeouts) and PDF range requests still return 206.

Supervised, single worker

A systemd unit with a 120 s stop timeout for the 1200 s simulation call. Live run state is in-process, so a worker count is structurally refused.

What it protects

Anonymous access to the API, the interface, the event stream, and the schema — the interactive docs are withdrawn whenever auth is on.

What it is not

One shared credential identifies nobody, rate-limits nothing, and caps no spend. The Funnel hostname is published to Certificate Transparency logs.

The credential is transport protection, not evidence of who did what—so exposure stays temporary and `reviewer\_id` stays self-declared.

Evaluation separates software correctness from scientific validity

Passing tests proves the platform behaves as declared; it does not prove a model represents a facility.

A

Automated verification

- Backend tests
- Frontend interaction tests
- Production build
- Environment doctor
- Model validation checks
- Raw integrity + study quality

N

Numerical regression

- Pinned upstream trajectories
- Normalized RMSE threshold $\leq 0.5\%$
- Benchmark repeatability
- Metric completeness
- Physical plausibility screens

H

Manual blind review

- Traceability
- Missing-evidence visibility
- Preregistration
- Reproducibility
- Calibration disclosure
- Verification time

Manual review uses at least two reviewers and never collapses into a single 'scientific validity score.'

Current state: a tested research prototype with a credible benchmark

The platform is ready for guided research use and PI scrutiny—not unsupervised scientific deployment.

174

backend tests passed

60

frontend tests passed

7 / 7

conceptual model checks

valid

18-case study quality

Strong today

Auditable workflow, bounded registry, pinned runtime, evidence hashes, revisioned analysis, coherent UI, and a supervised authenticated deployment.

Needs research work

Facility calibration, uncertainty quantification, broader scenarios, and independent replication.

Needs product work

Per-reviewer identity, rate limiting and spend caps, multi-user provenance, long-running job infrastructure, and observability.

Readiness claim: strong prototype for supervised, reproducible computational studies.

The largest risks are scientific, not cosmetic

Each risk has a concrete mitigation path; none should be hidden by a polished interface.

Calibration gap

Library regression is not facility validation.

MITIGATION

Acquire measured plant data; calibrate and hold out validation periods.

Scenario narrowness

One dry-cold weather profile and a 3×3 operating matrix.

MITIGATION

Add climates, seasons, load profiles, faults, and control variants.

LLM dependence

Hypothesis quality varies with model and retrieval coverage.

MITIGATION

Benchmark prompts/models; capture uncertainty; retain human selection.

Numerical events

Isolated mover-power impulses require a declared screen.

MITIGATION

Investigate upstream dynamics and compare alternate solvers/tolerances.

The platform makes these limitations easier to see; resolving them remains research work.

The next phase should deepen evidence before broadening automation

A staged roadmap keeps scientific credibility ahead of product surface area.

0-3 months

VALIDATE

- Replicate benchmark
- Investigate power impulses
- Add uncertainty bands
- Define calibration dataset

3-6 months

GENERALIZE

- New weather profiles
- Dynamic load traces
- Control variants
- Cross-solver comparison

6-12 months

TRANSLATE

- Facility calibration
- Prospective case study
- Independent reviewer study
- Operational deployment plan

Decision rule: do not add a new autonomous step until its inputs, outputs, failure modes, and review point are explicit.

Three decisions unlock the next research phase

01

Scientific target

Optimize energy only—or formalize a weighted energy / thermal-margin objective?

02

Validation access

Which facility data, collaborator, or measured benchmark can anchor calibration?

03

Evaluation design

Which users and reviewers should test traceability and verification-time gains?

Take-away

RCP can make simulation research faster and more reviewable—provided the model, protocol, and human judgment remain first-class evidence.